PART-1: GPT-2 Sentiment Analysis

과제 설명에 기재된 기준치입니다.

	SST (5-class)	CFIMDB (binary)
last-linear-layer	0.462	0.861
full-model	0.513	0.976

본 노트북에선 sanity_check.py, optimizer_test.py, classifier.py를 그대로 실행할 수 없어 로직의 변경 없이 함수를 호출해 실행하였습니다.

env 에 기재된 대로 conda를 설치하면 cpu 모드로 torch가 설치되어 같은 버전인 2.4.1의 cuda 버전으로 pip install을 실행하였습니다.
과제에서 정한 classifier.py 까지 실행한 뒤 마지막 섹션에서 시각화를 위해 matplotlib을 추가적으로 설치하였습니다.

In [9]:
```
from pathlib import Path
import sys, os

PROJECT_ROOT = Path.cwd()
if not (PROJECT_ROOT / 'classifier.py').exists():
    PROJECT_ROOT = PROJECT_ROOT.parent
os.chdir(str(PROJECT_ROOT))
sys.path.insert(0, str(PROJECT_ROOT))
print('PROJECT_ROOT:', PROJECT_ROOT)
```

```
PROJECT_ROOT: c:\Users\ssant\nlp2026-final
```

1. 구현 검증

In [6]:
```
from sanity_check import test_gpt2
test_gpt2('gpt2')
```

```
Your GPT2 implementation is correct!
```

In [7]:
```
import numpy as np
import torch
from optimizer_test import test_optimizer
from optimizer import AdamW

ref = torch.tensor(np.load('optimizer_test.npy'))
actual = test_optimizer(AdamW)
assert torch.allclose(ref, actual, atol=1e-6, rtol=1e-4)
print('Optimizer test passed!')
```

```
Optimizer test passed!
```

2. Fine-tuning

[SST / CFIMDB] 두 데이터셋과 각각에 대해 [last / full] 레이어를 학습시켜 총 4가지 학습을 진행합니다. classifier.py를 수정 없이 그대로 실행하였습니다.

In [8]:
```python
# SST - last-linear-layer
import importlib
import classifier
from types import SimpleNamespace

importlib.reload(classifier)
classifier.seed_everything(11711)

config = SimpleNamespace(
    filepath='sst-last-linear-classifier.pt',
    lr=1e-3, use_gpu=True, epochs=10, batch_size=64,
    hidden_dropout_prob=0.3,
    train='data/ids-sst-train.csv',
    dev='data/ids-sst-dev.csv',
    test='data/ids-sst-test-student.csv',
    fine_tune_mode='last-linear-layer',
    dev_out='predictions/last-linear-layer-sst-dev-out.csv',
    test_out='predictions/last-linear-layer-sst-test-out.csv',
)
classifier.train(config)
classifier.test(config)
```

```
load 8544 data from data/ids-sst-train.csv
load 1101 data from data/ids-sst-dev.csv
train-0: 100%|██████████| 134/134 [00:09<00:00, 13.43it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.60it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 14.50it/s]
train-1:   1%|▏         | 2/134 [00:00<00:09, 13.68it/s]
save the model to sst-last-linear-classifier.pt
Epoch 0: train loss :: 1.833, train acc :: 0.417, dev acc :: 0.421
train-1: 100%|██████████| 134/134 [00:09<00:00, 14.11it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.70it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.80it/s]
train-2:   1%|▏         | 2/134 [00:00<00:09, 13.41it/s]
save the model to sst-last-linear-classifier.pt
Epoch 1: train loss :: 1.431, train acc :: 0.432, dev acc :: 0.428
train-2: 100%|██████████| 134/134 [00:09<00:00, 14.14it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.55it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.82it/s]
train-3:   1%|▏         | 2/134 [00:00<00:09, 13.36it/s]
Epoch 2: train loss :: 1.395, train acc :: 0.385, dev acc :: 0.378
train-3: 100%|██████████| 134/134 [00:09<00:00, 14.12it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.58it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 14.50it/s]
train-4:   1%|▏         | 2/134 [00:00<00:08, 15.88it/s]
save the model to sst-last-linear-classifier.pt
Epoch 3: train loss :: 1.363, train acc :: 0.468, dev acc :: 0.446
train-4: 100%|██████████| 134/134 [00:09<00:00, 14.13it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.68it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.82it/s]
train-5:   1%|▏         | 2/134 [00:00<00:09, 14.53it/s]
Epoch 4: train loss :: 1.349, train acc :: 0.443, dev acc :: 0.427
```

```
train-5: 100%|██████████| 134/134 [00:09<00:00, 14.21it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.58it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.68it/s]
train-6:   1%|▏         | 2/134 [00:00<00:09, 14.35it/s]
Epoch 5: train loss :: 1.332, train acc :: 0.457, dev acc :: 0.434
train-6: 100%|██████████| 134/134 [00:09<00:00, 14.01it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.50it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.76it/s]
train-7:   1%|▏         | 2/134 [00:00<00:08, 15.54it/s]
Epoch 6: train loss :: 1.329, train acc :: 0.466, dev acc :: 0.440
train-7: 100%|██████████| 134/134 [00:09<00:00, 14.11it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.56it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.77it/s]
train-8:   1%|▏         | 2/134 [00:00<00:09, 13.70it/s]
Epoch 7: train loss :: 1.332, train acc :: 0.477, dev acc :: 0.441
train-8: 100%|██████████| 134/134 [00:09<00:00, 14.04it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.46it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.74it/s]
train-9:   1%|▏         | 2/134 [00:00<00:09, 14.38it/s]
Epoch 8: train loss :: 1.316, train acc :: 0.466, dev acc :: 0.443
train-9: 100%|██████████| 134/134 [00:09<00:00, 14.13it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.56it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.74it/s]
save the model to sst-last-linear-classifier.pt
Epoch 9: train loss :: 1.316, train acc :: 0.487, dev acc :: 0.452
load model from sst-last-linear-classifier.pt
load 1101 data from data/ids-sst-dev.csv
eval: 100%|██████████| 18/18 [00:01<00:00, 12.84it/s]
eval:   6%|▌         | 2/35 [00:00<00:02, 13.09it/s]
DONE DEV
eval: 100%|██████████| 35/35 [00:02<00:00, 14.04it/s]
DONE Test
dev acc :: 0.452
```

In [10]:

```
# SST - full-model
importlib.reload(classifier)
classifier.seed_everything(11711)

config = SimpleNamespace(
    filepath='sst-full-model-classifier.pt',
    lr=1e-5, use_gpu=True, epochs=10, batch_size=64,
    hidden_dropout_prob=0.3,
    train='data/ids-sst-train.csv',
    dev='data/ids-sst-dev.csv',
    test='data/ids-sst-test-student.csv',
    fine_tune_mode='full-model',
    dev_out='predictions/full-model-sst-dev-out.csv',
    test_out='predictions/full-model-sst-test-out.csv',
)
classifier.train(config)
classifier.test(config)
```

```
load 8544 data from data/ids-sst-train.csv
load 1101 data from data/ids-sst-dev.csv
train-0: 100%|██████████| 134/134 [00:31<00:00,  4.31it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.46it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 14.38it/s]
train-1:   0%|          | 0/134 [00:00<?, ?it/s]
```

```
save the model to sst-full-model-classifier.pt
Epoch 0: train loss :: 1.897, train acc :: 0.307, dev acc :: 0.290
train-1: 100%|██████████| 134/134 [00:30<00:00,  4.39it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.51it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.39it/s]
train-2:   0%|          | 0/134 [00:00<?, ?it/s]
save the model to sst-full-model-classifier.pt
Epoch 1: train loss :: 1.562, train acc :: 0.398, dev acc :: 0.381
train-2: 100%|██████████| 134/134 [00:30<00:00,  4.35it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.42it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.48it/s]
train-3:   0%|          | 0/134 [00:00<?, ?it/s]
save the model to sst-full-model-classifier.pt
Epoch 2: train loss :: 1.416, train acc :: 0.482, dev acc :: 0.431
train-3: 100%|██████████| 134/134 [00:30<00:00,  4.35it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.49it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.56it/s]
train-4:   1%|          | 1/134 [00:00<00:25,  5.14it/s]
save the model to sst-full-model-classifier.pt
Epoch 3: train loss :: 1.241, train acc :: 0.531, dev acc :: 0.474
train-4: 100%|██████████| 134/134 [00:30<00:00,  4.37it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.41it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.66it/s]
train-5:   0%|          | 0/134 [00:00<?, ?it/s]
save the model to sst-full-model-classifier.pt
Epoch 4: train loss :: 1.170, train acc :: 0.560, dev acc :: 0.500
train-5: 100%|██████████| 134/134 [00:30<00:00,  4.37it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.32it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.54it/s]
train-6:   0%|          | 0/134 [00:00<?, ?it/s]
Epoch 5: train loss :: 1.120, train acc :: 0.564, dev acc :: 0.483
train-6: 100%|██████████| 134/134 [00:30<00:00,  4.33it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.37it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.68it/s]
train-7:   0%|          | 0/134 [00:00<?, ?it/s]
Epoch 6: train loss :: 1.071, train acc :: 0.594, dev acc :: 0.495
train-7: 100%|██████████| 134/134 [00:30<00:00,  4.37it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.43it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.64it/s]
train-8:   0%|          | 0/134 [00:00<?, ?it/s]
Epoch 7: train loss :: 1.036, train acc :: 0.612, dev acc :: 0.496
train-8: 100%|██████████| 134/134 [00:30<00:00,  4.35it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.30it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.53it/s]
train-9:   0%|          | 0/134 [00:00<?, ?it/s]
Epoch 8: train loss :: 1.001, train acc :: 0.623, dev acc :: 0.493
train-9: 100%|██████████| 134/134 [00:30<00:00,  4.36it/s]
eval: 100%|██████████| 134/134 [00:09<00:00, 14.33it/s]
eval: 100%|██████████| 18/18 [00:01<00:00, 15.50it/s]
Epoch 9: train loss :: 0.966, train acc :: 0.646, dev acc :: 0.496
load model from sst-full-model-classifier.pt
load 1101 data from data/ids-sst-dev.csv
eval: 100%|██████████| 18/18 [00:01<00:00, 13.28it/s]
eval:   6%|▌         | 2/35 [00:00<00:02, 13.12it/s]
DONE DEV
eval: 100%|██████████| 35/35 [00:02<00:00, 13.94it/s]
```

```
DONE Test
dev acc :: 0.500
```

In [12]:

```python
# CFIMDB - last-linear-layer
importlib.reload(classifier)
classifier.seed_everything(11711)

config = SimpleNamespace(
    filepath='cfimdb-last-linear-classifier.pt',
    lr=1e-3, use_gpu=True, epochs=10, batch_size=8,
    hidden_dropout_prob=0.3,
    train='data/ids-cfimdb-train.csv',
    dev='data/ids-cfimdb-dev.csv',
    test='data/ids-cfimdb-test-student.csv',
    fine_tune_mode='last-linear-layer',
    dev_out='predictions/last-linear-layer-cfimdb-dev-out.csv',
    test_out='predictions/last-linear-layer-cfimdb-test-out.csv',
)
classifier.train(config)
classifier.test(config)
```

```
load 1707 data from data/ids-cfimdb-train.csv
load 245 data from data/ids-cfimdb-dev.csv
train-0: 100%|██████████| 214/214 [00:21<00:00,  9.84it/s]
eval: 100%|██████████| 214/214 [00:19<00:00, 10.92it/s]
eval: 100%|██████████| 31/31 [00:02<00:00, 10.47it/s]
train-1:   0%|          | 1/214 [00:00<00:26,  7.99it/s]
save the model to cfimdb-last-linear-classifier.pt
Epoch 0: train loss :: 0.701, train acc :: 0.670, dev acc :: 0.649
train-1: 100%|██████████| 214/214 [00:21<00:00, 10.18it/s]
eval: 100%|██████████| 214/214 [00:19<00:00, 10.92it/s]
eval: 100%|██████████| 31/31 [00:02<00:00, 10.98it/s]
train-2:   1%|          | 2/214 [00:00<00:13, 15.58it/s]
save the model to cfimdb-last-linear-classifier.pt
Epoch 1: train loss :: 0.564, train acc :: 0.820, dev acc :: 0.800
train-2: 100%|██████████| 214/214 [00:21<00:00, 10.00it/s]
eval: 100%|██████████| 214/214 [00:19<00:00, 10.92it/s]
eval: 100%|██████████| 31/31 [00:02<00:00, 10.94it/s]
train-3:   0%|          | 0/214 [00:00<?, ?it/s]
Epoch 2: train loss :: 0.510, train acc :: 0.813, dev acc :: 0.800
train-3: 100%|██████████| 214/214 [00:21<00:00,  9.80it/s]
eval: 100%|██████████| 214/214 [00:19<00:00, 11.02it/s]
eval: 100%|██████████| 31/31 [00:02<00:00, 11.12it/s]
train-4:   0%|          | 1/214 [00:00<00:27,  7.83it/s]
save the model to cfimdb-last-linear-classifier.pt
Epoch 3: train loss :: 0.482, train acc :: 0.865, dev acc :: 0.837
train-4: 100%|██████████| 214/214 [00:21<00:00, 10.07it/s]
eval: 100%|██████████| 214/214 [00:19<00:00, 11.19it/s]
eval: 100%|██████████| 31/31 [00:02<00:00, 11.10it/s]
train-5:   1%|          | 2/214 [00:00<00:14, 15.05it/s]
Epoch 4: train loss :: 0.479, train acc :: 0.858, dev acc :: 0.820
train-5: 100%|██████████| 214/214 [00:20<00:00, 10.47it/s]
eval: 100%|██████████| 214/214 [00:19<00:00, 11.09it/s]
eval: 100%|██████████| 31/31 [00:02<00:00, 11.09it/s]
train-6:   1%|          | 2/214 [00:00<00:17, 11.79it/s]
Epoch 5: train loss :: 0.483, train acc :: 0.865, dev acc :: 0.837
```

```
train-6: 100%|██████████| 214/214 [00:21<00:00, 10.13it/s]
eval: 100%|██████████| 214/214 [00:19<00:00, 11.05it/s]
eval: 100%|██████████| 31/31 [00:02<00:00, 11.08it/s]
train-7:   0%|          | 1/214 [00:00<00:26,  7.98it/s]
Epoch 6: train loss :: 0.484, train acc :: 0.800, dev acc :: 0.792
train-7: 100%|██████████| 214/214 [00:20<00:00, 10.24it/s]
eval: 100%|██████████| 214/214 [00:18<00:00, 11.27it/s]
eval: 100%|██████████| 31/31 [00:02<00:00, 11.07it/s]
train-8:   1%|          | 2/214 [00:00<00:16, 13.11it/s]
Epoch 7: train loss :: 0.448, train acc :: 0.844, dev acc :: 0.784
train-8: 100%|██████████| 214/214 [00:20<00:00, 10.28it/s]
eval: 100%|██████████| 214/214 [00:19<00:00, 11.21it/s]
eval: 100%|██████████| 31/31 [00:02<00:00, 11.12it/s]
train-9:   1%|          | 2/214 [00:00<00:16, 12.51it/s]
Epoch 8: train loss :: 0.462, train acc :: 0.851, dev acc :: 0.804
train-9: 100%|██████████| 214/214 [00:21<00:00, 10.18it/s]
eval: 100%|██████████| 214/214 [00:18<00:00, 11.29it/s]
eval: 100%|██████████| 31/31 [00:02<00:00, 11.11it/s]
save the model to cfimdb-last-linear-classifier.pt
Epoch 9: train loss :: 0.433, train acc :: 0.877, dev acc :: 0.865
load model from cfimdb-last-linear-classifier.pt
load 245 data from data/ids-cfimdb-dev.csv
eval: 100%|██████████| 31/31 [00:03<00:00,  9.96it/s]
eval:   2%|▏         | 1/61 [00:00<00:07,  8.46it/s]
DONE DEV
eval: 100%|██████████| 61/61 [00:05<00:00, 10.40it/s]
DONE Test
dev acc :: 0.865
```

In [13]:

```python
# CFIMDB - full-model
importlib.reload(classifier)
classifier.seed_everything(11711)

config = SimpleNamespace(
    filepath='cfimdb-full-model-classifier.pt',
    lr=1e-5, use_gpu=True, epochs=10, batch_size=8,
    hidden_dropout_prob=0.3,
    train='data/ids-cfimdb-train.csv',
    dev='data/ids-cfimdb-dev.csv',
    test='data/ids-cfimdb-test-student.csv',
    fine_tune_mode='full-model',
    dev_out='predictions/full-model-cfimdb-dev-out.csv',
    test_out='predictions/full-model-cfimdb-test-out.csv',
)
classifier.train(config)
classifier.test(config)
```

```
load 1707 data from data/ids-cfimdb-train.csv
load 245 data from data/ids-cfimdb-dev.csv
train-0: 100%|██████████| 214/214 [04:56<00:00,  1.39s/it]
eval: 100%|██████████| 214/214 [01:27<00:00,  2.46it/s]
eval: 100%|██████████| 31/31 [00:16<00:00,  1.83it/s]
train-1:   0%|          | 0/214 [00:00<?, ?it/s]
save the model to cfimdb-full-model-classifier.pt
Epoch 0: train loss :: 0.616, train acc :: 0.965, dev acc :: 0.967
```

```
train-1: 100%|██████████| 214/214 [05:50<00:00,  1.64s/it]
eval: 100%|██████████| 214/214 [01:46<00:00,  2.02it/s]
eval: 100%|██████████| 31/31 [00:20<00:00,  1.49it/s]
train-2:   0%|          | 0/214 [00:00<?, ?it/s]
save the model to cfimdb-full-model-classifier.pt
Epoch 1: train loss :: 0.142, train acc :: 0.987, dev acc :: 0.976
train-2: 100%|██████████| 214/214 [05:57<00:00,  1.67s/it]
eval: 100%|██████████| 214/214 [01:36<00:00,  2.23it/s]
eval: 100%|██████████| 31/31 [00:14<00:00,  2.15it/s]
train-3:   0%|          | 0/214 [00:00<?, ?it/s]
Epoch 2: train loss :: 0.085, train acc :: 0.989, dev acc :: 0.976
train-3: 100%|██████████| 214/214 [03:28<00:00,  1.03it/s]
eval: 100%|██████████| 214/214 [01:04<00:00,  3.31it/s]
eval: 100%|██████████| 31/31 [00:11<00:00,  2.61it/s]
train-4:   0%|          | 0/214 [00:00<?, ?it/s]
save the model to cfimdb-full-model-classifier.pt
Epoch 3: train loss :: 0.055, train acc :: 0.995, dev acc :: 0.980
train-4: 100%|██████████| 214/214 [04:19<00:00,  1.21s/it]
eval: 100%|██████████| 214/214 [01:20<00:00,  2.66it/s]
eval: 100%|██████████| 31/31 [00:15<00:00,  2.02it/s]
train-5:   0%|          | 0/214 [00:00<?, ?it/s]
Epoch 4: train loss :: 0.042, train acc :: 0.996, dev acc :: 0.951
train-5: 100%|██████████| 214/214 [04:49<00:00,  1.35s/it]
eval: 100%|██████████| 214/214 [01:29<00:00,  2.39it/s]
eval: 100%|██████████| 31/31 [00:13<00:00,  2.31it/s]
train-6:   0%|          | 0/214 [00:00<?, ?it/s]
Epoch 5: train loss :: 0.033, train acc :: 0.996, dev acc :: 0.976
train-6: 100%|██████████| 214/214 [04:25<00:00,  1.24s/it]
eval: 100%|██████████| 214/214 [01:20<00:00,  2.67it/s]
eval: 100%|██████████| 31/31 [00:16<00:00,  1.91it/s]
train-7:   0%|          | 0/214 [00:00<?, ?it/s]
save the model to cfimdb-full-model-classifier.pt
Epoch 6: train loss :: 0.019, train acc :: 0.999, dev acc :: 0.984
train-7: 100%|██████████| 214/214 [04:32<00:00,  1.27s/it]
eval: 100%|██████████| 214/214 [00:59<00:00,  3.60it/s]
eval: 100%|██████████| 31/31 [00:11<00:00,  2.60it/s]
train-8:   0%|          | 0/214 [00:00<?, ?it/s]
Epoch 7: train loss :: 0.026, train acc :: 0.998, dev acc :: 0.976
train-8: 100%|██████████| 214/214 [03:28<00:00,  1.03it/s]
eval: 100%|██████████| 214/214 [01:03<00:00,  3.37it/s]
eval: 100%|██████████| 31/31 [00:11<00:00,  2.64it/s]
train-9:   0%|          | 0/214 [00:00<?, ?it/s]
Epoch 8: train loss :: 0.009, train acc :: 0.998, dev acc :: 0.967
train-9: 100%|██████████| 214/214 [03:38<00:00,  1.02s/it]
eval: 100%|██████████| 214/214 [01:03<00:00,  3.37it/s]
eval: 100%|██████████| 31/31 [00:11<00:00,  2.60it/s]
Epoch 9: train loss :: 0.011, train acc :: 0.996, dev acc :: 0.984
load model from cfimdb-full-model-classifier.pt
load 245 data from data/ids-cfimdb-dev.csv
eval: 100%|██████████| 31/31 [00:12<00:00,  2.57it/s]
eval:   0%|          | 0/61 [00:00<?, ?it/s]
DONE DEV
eval: 100%|██████████| 61/61 [00:21<00:00,  2.79it/s]
DONE Test
dev acc :: 0.984
```

3. 결과 요약

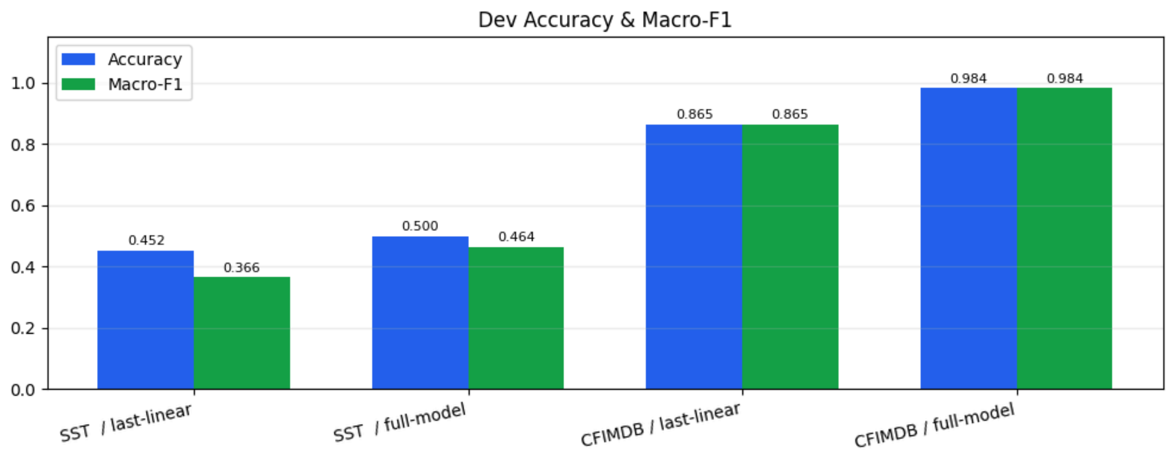
In [14]:
```
import csv
from sklearn.metrics import accuracy_score, f1_score, confusion_matrix, precisio

EVAL_CONFIGS = [
    ('SST  / last-linear', 'data/ids-sst-dev.csv',     'predictions/last-linear-
    ('SST  / full-model',  'data/ids-sst-dev.csv',     'predictions/full-model-s
    ('CFIMDB / last-linear', 'data/ids-cfimdb-dev.csv', 'predictions/last-linear
    ('CFIMDB / full-model',  'data/ids-cfimdb-dev.csv', 'predictions/full-model-
]

def read_dev(path):
    with open(path, newline='') as f:
        return {r['id'].strip(): int(r['sentiment']) for r in csv.DictReader(f,

def read_pred(path):
    with open(path, newline='') as f:
        return {r['id'].strip(): int(r['Predicted_Sentiment']) for r in csv.Dict

results = []
print(f"{'Config':<25} {'Accuracy':>10} {'Macro-F1':>10}")
print('-' * 48)
for label, dev_path, pred_path in EVAL_CONFIGS:
    if not Path(pred_path).exists():
        print(f"{label:<25} {'(no file)':>10}")
        continue
    true = read_dev(dev_path)
    pred = read_pred(pred_path)
    ids  = [k for k in pred if k in true]
    y_true = [true[k] for k in ids]
    y_pred = [pred[k]  for k in ids]
    acc = accuracy_score(y_true, y_pred)
    f1  = f1_score(y_true, y_pred, average='macro')
    results.append({'label': label, 'acc': acc, 'f1': f1, 'y_true': y_true, 'y_p
    print(f"{label:<25} {acc:>10.4f} {f1:>10.4f}")
```

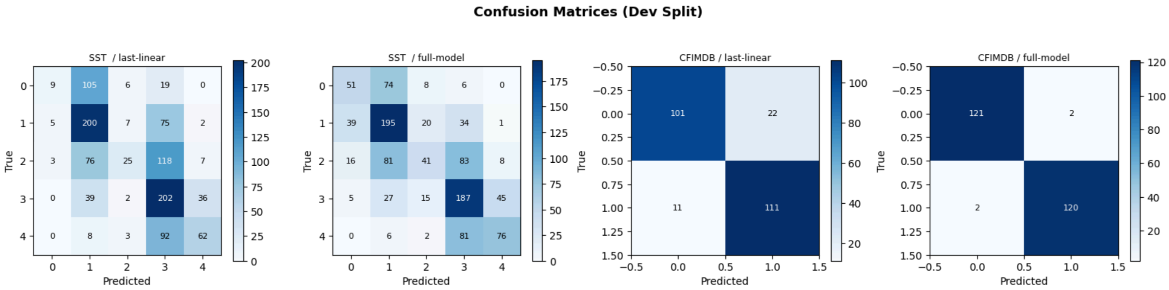
```
Config                      Accuracy   Macro-F1
------------------------------------------------
SST  / last-linear            0.4523     0.3655
SST  / full-model             0.4995     0.4644
CFIMDB / last-linear          0.8653     0.8651
CFIMDB / full-model           0.9837     0.9837
```

4. 시각화 & Confusion Matrix

In [15]:
```
import matplotlib.pyplot as plt
import numpy as np
from sklearn.metrics import confusion_matrix

if not results:
    print('결과 없음 — Section 3 셀을 먼저 실행하세요.')
else:
    labels = [r['label'] for r in results]
    accs   = [r['acc']   for r in results]
    f1s    = [r['f1']    for r in results]
    x = np.arange(len(labels))
```

```python
w = 0.35

# ── Accuracy & Macro-F1 bar chart ────────────────────────
fig, ax = plt.subplots(figsize=(10, 4))
b1 = ax.bar(x - w/2, accs, w, label='Accuracy', color='#2563eb')
b2 = ax.bar(x + w/2, f1s,  w, label='Macro-F1', color='#16a34a')
ax.set_xticks(x)
ax.set_xticklabels(labels, rotation=12, ha='right')
ax.set_ylim(0, 1.15)
ax.set_title('Dev Accuracy & Macro-F1')
ax.legend()
ax.grid(axis='y', alpha=0.25)
for bar, val in zip(list(b1) + list(b2), accs + f1s):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.01,
            f'{val:.3f}', ha='center', va='bottom', fontsize=8)
plt.tight_layout()
plt.show()

# ── Confusion matrices ───────────────────────────────────
n   = len(results)
fig, axes = plt.subplots(1, n, figsize=(4 * n, 4))
if n == 1:
    axes = [axes]
fig.suptitle('Confusion Matrices (Dev Split)', fontsize=13, fontweight='bold
for ax, r in zip(axes, results):
    cm     = confusion_matrix(r['y_true'], r['y_pred'])
    im     = ax.imshow(cm, cmap='Blues')
    thresh = cm.max() / 2
    ax.set_title(r['label'], fontsize=9)
    ax.set_xlabel('Predicted')
    ax.set_ylabel('True')
    for i in range(cm.shape[0]):
        for j in range(cm.shape[1]):
            ax.text(j, i, cm[i, j], ha='center', va='center',
                    color='white' if cm[i, j] > thresh else 'black', fontsiz
    plt.colorbar(im, ax=ax, shrink=0.8)
plt.tight_layout()
plt.show()
```




Confusion Matrices (Dev Split)
SST / last-linear
SST / full-model
CFIMDB / last-linear
CFIMDB / full-model
True
Predicted